



STREETLIGHT

All Vehicles Aggregated GPS Volume Methodology and Validation White Paper

November 2025

Version 3.0

Table of Contents

Introduction.....	3
Methodology	3
Aggregating Proprietary Data	3
Analysis Support	4
Vehicle Volume Overview	4
Vehicle Volume Algorithm	4
Input Features	5
Ground-Truth Data.....	5
Filling Coverage Gaps	6
Algorithm Details.....	7
Volume Prediction Intervals	7
Estimating Volume from the MADT Model for Specific Time Intervals.....	7
Validation	8
Vehicle Volume Validation Methodology.....	8
Validation Results for Vehicle Volume	8
Comparability Across Time	12
Limitations	13
Conclusions and Next Steps.....	13
About StreetLight Data.....	13

Introduction

StreetLight Data, Inc. (“StreetLight”) pioneered the use of Big Data analytics to shed light on how people, goods, and services move, empowering smarter, data-driven transportation decisions. StreetLight's proprietary data processing engine, Route Science®, algorithmically transforms its vast data resources to measure travel patterns of vehicles, bicycles and pedestrians. StreetLight provides innovative digital solutions to help communities reduce congestion, improve safe and equitable transportation, and maximize the positive impact of infrastructure investment.

This white paper describes the data sources and methodology employed by StreetLight to develop Vehicle Volume Metrics.

Methodology

To generate StreetLight metrics, we start with sample trips and then build models to combine these trip counts with contextual information, such as demographics, road characteristics, weather, and seasonality. Estimated volume metrics are validated against traffic data collected by state and regional agencies.

Aggregating Proprietary Data

StreetLight's All Vehicles Volume metrics for certain analysis types are powered by Aggregated GPS (AGPS) Data. This includes a unique blend of device navigation apps, traditional mobile data apps, and in-vehicle navigation apps. This source is innately different from Location Based Services (LBS) data and Connected Vehicle Data (CVD) sources utilized in other StreetLight products and StreetLight InSight® analyses.

Aggregated GPS Data provides some advantages in that it has a highly representative sample with strong penetration rates across road sizes and regions. Penetration rates are measured as sample trip counts divided by ground-truth vehicle volumes collected at permanent counter locations. In an evaluation of penetration rates across U.S. roads in 2025, we found that on average, our Aggregated GPS Data source captured 34.2% of vehicles on the road. This extremely large sample contributes to strong metrics quality, especially on rural and lower-volume roads.

When leveraging AGPS Data for Vehicle Volume, we proactively source and store metrics for all road segments in the U.S. at the 15-minute level. This approach allows us to quickly access relevant data for any selected road segment without additional trip processing or data aggregation steps. As with any of our data sources, this data undergoes thorough quality

assurance to ensure that no errors have occurred in data ingestion and that data coverage is spatially and temporally consistent.

Analysis Support

All Vehicles Volumes derived from AGPS data are used in our segment-level analysis types within StreetLight InSight®, including Network Performance, Network Origin-Destination, and Turning Movement Counts. All Vehicles Volumes based on AGPS data are distinct from All Vehicles CVD+ Volumes, which are used in our non-pass-through analyses in StreetLight InSight®, such as Zone Activity, Origin-Destination, and Top Routes. To read more about the CVD Volumes methodology and validation, [see this white paper](#).

Vehicle Volume Overview

To develop segment-level metrics, StreetLight created an algorithm that maps segments from the segment library of our AGPS data provider to the StreetLight InSight® Zone Library, derived from OpenStreetMap (OSM). This allows for a seamless integration of vehicle trip data traversing segments into StreetLight's metrics.

Once the data has been ingested, StreetLight employs multiple algorithms including machine-learning models to estimate StreetLight volume, as well as metrics like speed, travel time, etc. When a customer selects a road segment or series of road segments for analysis, StreetLight extracts relevant data from segments based on the customer's desired data period, day types, and day parts.

In the following sections, we use the terms "Volume" and "MADT". Volume refers to traffic volume generally and can apply to any time period. MADT (Monthly Average Daily Traffic) is a more specific metric that refers to the average daily traffic volume in a given month.

Vehicle Volume Algorithm

This is the third version of our AGPS volume algorithm. Version 3 improves upon Version 2 in the following ways:

- Near-100% coverage of residential roads and other low-volume road classes, across all years.
- From 2023 onwards, the V3 model leverages an updated segment library from our AGPS data provider, which helps improve accuracy and helps reduce segment mapping issues like empty results or mapping irregularities.

- Improved bias, meaning we are less likely to generally estimate volumes to be higher or lower than counter values.
- The addition of permanent counters used for training and validation from five more states each month, for 2025 data and for future monthly data updates.
- The inclusion of counters from all lower 48 states for 2024.

INPUT FEATURES

The following features are used as inputs in the volume model:

- **Sample trips:** Our processing algorithms obtain the trip count for a given road segment. This trip count represents all vehicle trips in the sample from our Aggregated GPS data provider.
- **Trip attributes:** Trip sample attributes such as average speed provide additional signal to the model.
- **U.S. Census data:** We use demographic data, such as number of housing units and whether or not a location is in an urban area.
- **OpenStreetMap (OSM) data:** OSM provides information on road classification, number of lanes, speed limit, and other attributes.
- **Seasonality:** Historical travel patterns are used to quantify the seasonality of a given area.
- **Region, year, and month:** Features representing the region, year and month are used to help the model further adjust for sample bias across space and time.

GROUND-TRUTH DATA

The volume model is trained and validated using data processed from over 14,000 permanent counters, representing more than 7,000 unique locations across the U.S. (Note that each location may have counters measuring more than one direction.) As shown in Figure 1, these counters cover all states in the contiguous U.S.

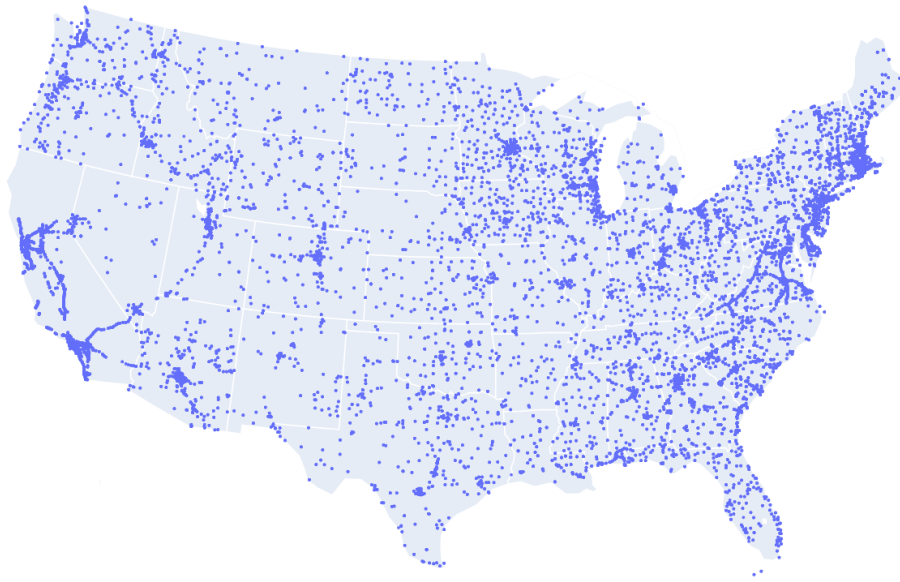


Figure 1: Locations of counters used in the vehicle volume model.

The model is trained using data from 2019 through 2025. At the time of model development, counter data from all states was not yet available for 2025. Thus, the 2025 dataset represents 27 states instead of all 48, as shown in Table 1. The 27 states includes five additional states that we employ each month in volume model training and validation. There may be three counter available at each location (two one-way and one bidirectional counter).

Year of Data	Number of States	Number of Locations	Number of Counters
2019	48	7,114	18,832
2020	48	7,198	19,067
2021	48	7,429	19,769
2022	48	7,778	20,620
2023	48	7,673	20,597
2024	48	7,783	20,837
2025	27	4,294	12,209

Table 1: Statistics on counters and counter location included in this model version, by year.

FILLING COVERAGE GAPS

With this model, we implemented two changes to fill gaps in coverage, especially on lower-volume roads such as residential roads.

First, we rolled out a new back-end road network map that reduced misconnections between our sample trip data and the OSM road segments that we provide volume metrics on. This update applies to months in 2023 onward. Previously, misconnections between the two maps might

have resulted in segments with reduced or missing data. In testing, this new segment library from our AGPS provider resulted in improved results for select segments across all road classes.

Second, our AGPS data does not consistently provide coverage on some rural and low volume roads. To improve our coverage, we inferred sample trip counts on small roads that previously had low sample, or were missing data. Specifically, we employed our large dataset of sample trip counts from similar roads to infer the trip counts on those missing roads. This helps us fill coverage gaps on residential, tertiary, and other small road classes. The inferred sample count is then passed as a feature into the volume model, described below. We verified that this method produced reasonable sample estimations and time trends. Roads that contain this additional imputation step will be tagged as “inferred volume” in analysis visuals and downloads.

ALGORITHM DETAILS

Once the input features and ground-truth data are processed, they are used to train an Extreme Gradient Boosting (XGBoost) model. The model estimates the trip sample penetration rate for a given location and time period, and that rate is then used to estimate vehicle volume. The algorithm also includes guardrails to protect against extreme outliers.

VOLUME PREDICTION INTERVALS

StreetLight volume estimates include prediction intervals that represent the likely upper and lower bounds of the true value. The 95% prediction interval is an estimate of the range within which the true value is expected to lie 95% of the time. To construct these intervals, we fit a quantile regression model to the MADT errors. The prediction intervals included with StreetLight’s volume metrics are only available for All Day day part and All Days day type estimates (not for specific day types or day parts) and only when there are at least 25 days in the analysis data period.

ESTIMATING VOLUME FROM THE MADT MODEL FOR SPECIFIC TIME INTERVALS

The volume models described in the previous section estimate monthly average daily traffic volumes (MADT) for weekday and weekend time periods. To generate volume estimates for day parts or day types, we use MADT as the base, then apply a time-specific scaling factor. The scaling factor is equal to the ratio of trip count for that time interval to trip count for that month. Our assumption here is that the ratio of sample trips to actual trips for a location is stable throughout the month. Possible trip sample bias by day part or day type is not presently accounted for in our volume metric.

Validation

Vehicle Volume Validation Methodology

To validate the MADT model, we used stratified k-fold cross-validation (CV) with $k = 10$. Cross-validation is primarily used to assess how well the model can predict monthly traffic volumes at permanent counter locations withheld from the training set. K-fold CV was a good choice in this case, where we have a large amount of training data, because it reduces bias and minimizes overfitting without being too computationally intensive. In performing the CV, we stratified by road volume category combined with their state (e.g. low-volume and California) to ensure minimal bias in training and testing the CV folds, and better estimate the model's ability to generalize on unseen sites.

We use the following set of error metrics to evaluate estimation accuracy:

- **Mean absolute percentage error (MAPE):** A standard measure of error for predictive machine-learning models. It describes errors well on small roads and treats errors of all sizes equally.
- **Normalized root mean square error (NRMSE):** Another standard error metric in machine learning. NRMSE penalizes large errors more, making it more sensitive to the accuracy of MADT estimation on high-volume roads.
- **68th percentile absolute error:** This figure represents one standard deviation from the mean within a standard bell curve and thus describes the typical error across the road segments. The 68th percentile absolute error can be interpreted as the expected typical error.
- **95th percentile absolute error:** The upper limit of the expected absolute error, or approximately two standard deviations from the mean.
- **Median percentage error:** A measure of bias in our model that also indicates how accurate an aggregate value (such as the sum of Volume on all roads in this bin) will be. Values close to zero suggest that our model has low bias, while positive values indicate more overestimation and negative values indicate more underestimation.

Validation Results for Vehicle Volume

In this section, we report cross-validation results for the vehicle volume model. As shown in Figure 2, the correlation between estimated MADT and permanent counter MADT is very strong. The correlation plot contains counter data from 2019-2025 months in aggregate. When evaluated annually, results are similarly strong, suggesting consistency in correlation at the yearly level.

Correlation between Counter MADT and Estimated MADT

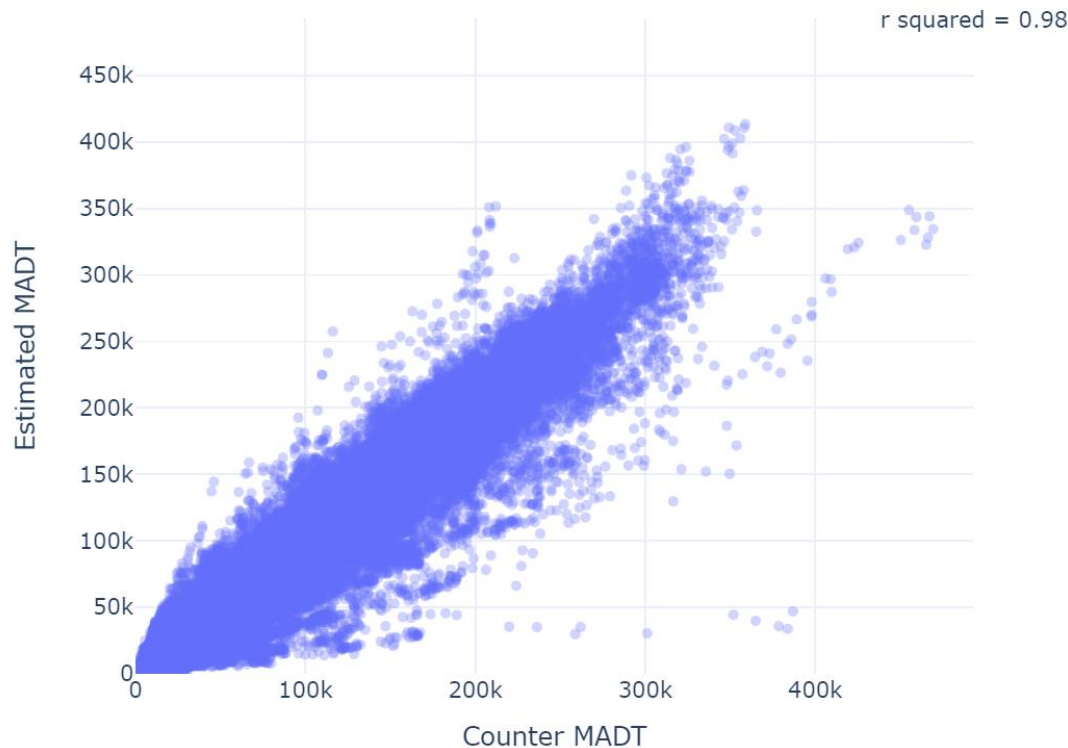


Figure 2: Correlation between StreetLight estimated MADT and counter MADT for 2019-2025. The best fit line has an R^2 value of 0.98. This is consistent year over year.

Table 2 shows the cross-validation results for the monthly volume model for all months from January 2019-July 2025, broken down by category of road volume. In general, errors decrease with larger road size. This is expected with errors measured in percentage terms, since higher volume roads have a larger denominator.

The results show minimal bias, which means the model is not generally estimating 'low' or 'high' in a substantial way. Absolute bias is 3% or less on all road sizes, even those with less than 500 in average daily traffic. On roads with more than 500 in average daily traffic, bias is less than 1%, only up to 2% on the largest roads. These metrics improve over the previous model version, which had larger bias (ranging from -1.6% to -2.9%) on all but the smallest roads.

Overall, MAPE is lower than 23.5% on all road sizes, and under 9% on high-volume roads with more than 35,000 in MADT.

The 68th percentile errors range from 16-26% for low volume roads to under 10% for larger roads. For example, road size F (20,000-34,999 MADT) has a 68th percentile error of 9.9%, indicating that the volume error for a typical road with about 25,000 in daily traffic would be about 10%, or +/- 2,500 as an absolute number.

The 95th percentile errors, which represent the extreme high and low end of expected errors, range from about 63% on the lowest volume roads to under 25% on the highest volume roads.

Road Size Category (based on MADT)	# of Counter Months	Median Percent Error (Bias) (%)	MAPE (%)	68th Abs Percentile Error (%)	95th Abs Percentile Error (%)	NRMSE (%)
A: 0 – 499	36,887	3.1	23.3	27.0	63.1	31.6
B: 500 – 1,999	153,334	0.7	16.0	18.4	44.3	22.4
C: 2,000 – 4,999	199,939	-0.5	13.3	15.3	36.7	18.8
D: 5,000 – 9,999	195,037	-0.3	11.7	13.5	31.6	16.4
E: 10,000 – 19,999	218,844	-0.2	10.1	11.4	27.6	14.7
F: 20,000 – 34,999	169,260	-0.2	9.1	9.9	26.1	13.8
G: 35,000 – 54,999	119,403	-0.5	8.9	9.4	26.6	13.5
H: 55,000 – 84,999	111,231	-1.0	8.2	8.7	24.9	12.7
I: 85,000 – 124,999	60,182	-1.8	8.5	9.2	24.6	12.5
J: 125,000 +	45,277	-2.0	8.3	8.9	24.4	13.1

Table 2: Cross-validation results for the vehicle volume model by road size. The metrics cover all months from January 2019-July 2025. Road volume category is based on the measured average MADT of the counter. Number of counters months refers to unique counter locations per month; each counter location can have multiple data points for different months/years and travel directions.

For a view of accuracy over time, Figures 3 and 4 show error metrics by month and road size category. MAPE (Figure 3) improves over time with the highest errors observed in 2019-2021. This is due to some volatility in the underlying AGPS sample during this time, and abnormal travel patterns during the Covid-19 pandemic. Bias (Figure 4) shows a similar pattern, with relatively consistent results from mid-2021 onwards.

The stability in error metrics over time also suggests that the model produces reliable time trends. Time trends are discussed in more detail in the following section.

MAPE by Month and Road Size

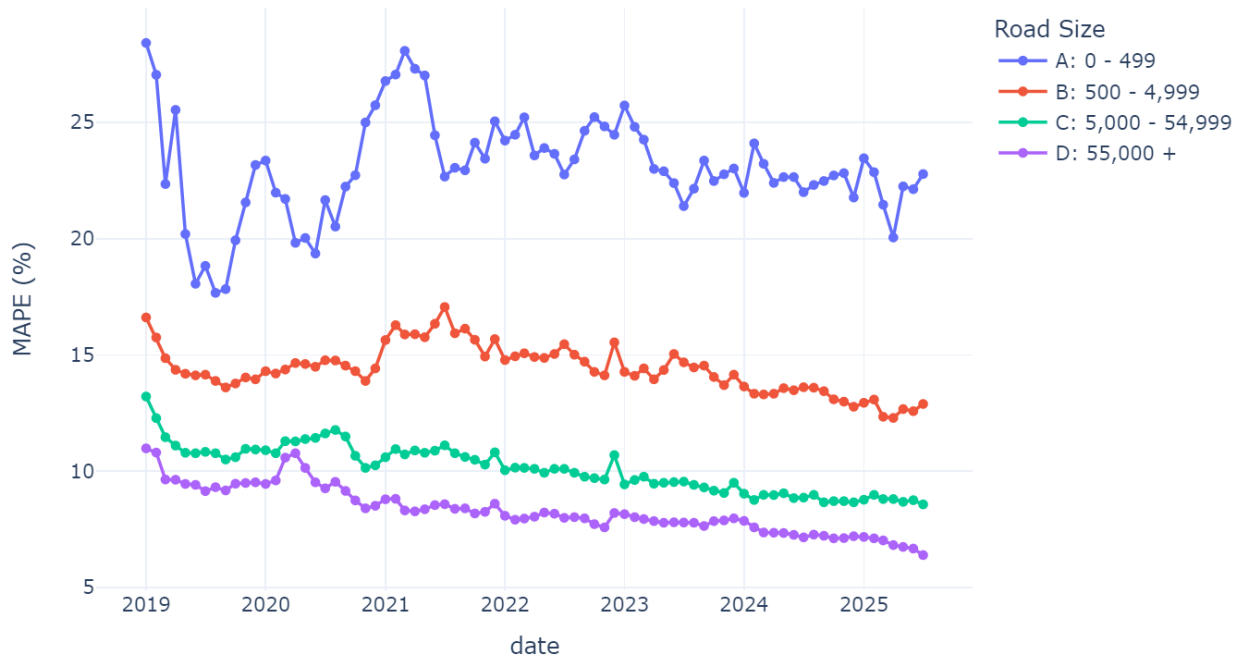


Figure 3: MAPE for StreetLight volume by month and road size, for all counters.

Median Percent Error (Bias) by Month and Road Size

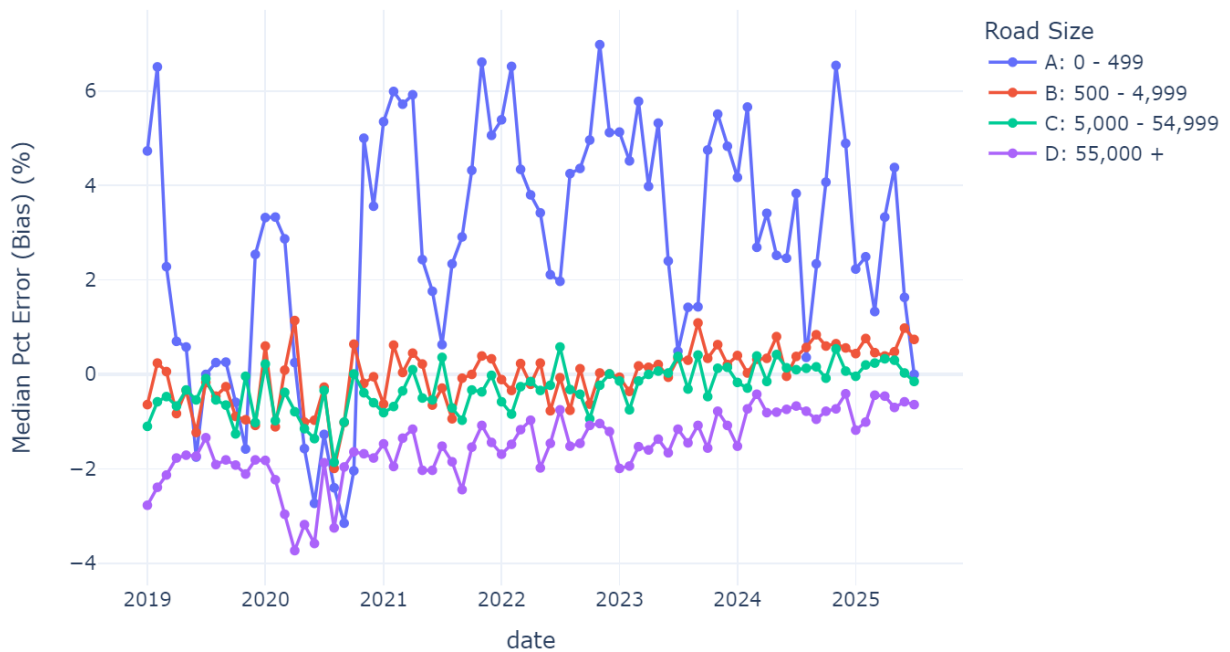


Figure 4: Median percent error (bias) for StreetLight volume by month and road size, for all counters.

COMPARABILITY ACROSS TIME

To evaluate the ability of the model to capture month-over-month time trends, we visually inspected results for a random sample of approximately 100 counter locations. Ten of these locations, one for each road size bin, are shown in Figure 5 below.

Estimated MADT and Counter MADT time trends for a sample of counters

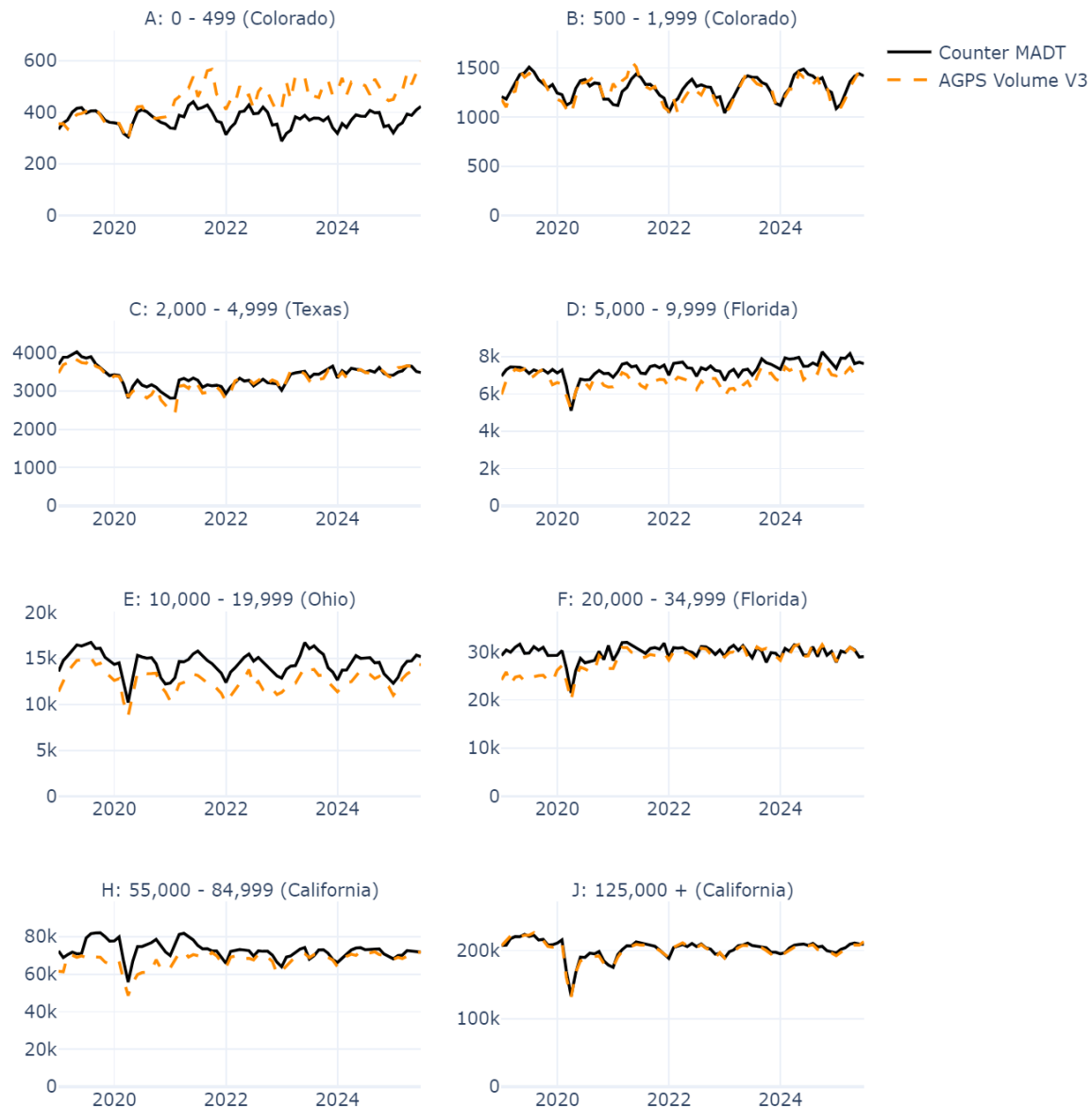


Figure 5: Estimated MADT compared to permanent counter MADT by month for a random selection of counters, stratified by road size.

Limitations

The errors presented in this paper are for locations with permanent counters, and even though we have good coverage across regions and road sizes, we do not have visibility into errors in places where permanent counters are not present. Additionally, due to some volatility in the AGPS sample in early 2019, errors may be slightly higher in this time frame relative to later months and years. Finally, due to fewer available counters in 2025 data months, errors may be slightly higher in this time frame as well. We plan to integrate additional permanent counters into our training set as they become available in the future.

Conclusions and Next Steps

At StreetLight, we are continually updating our data sources and algorithms to enhance the comprehensiveness and coverage of our metrics. The volume validations give us confidence in these metrics while also demonstrating where we can concentrate effort in further advances. Future iterations of our metrics will focus on improving accuracy across the full range of road sizes, while reducing outliers at the extreme ends of the error distribution.

About StreetLight Data

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